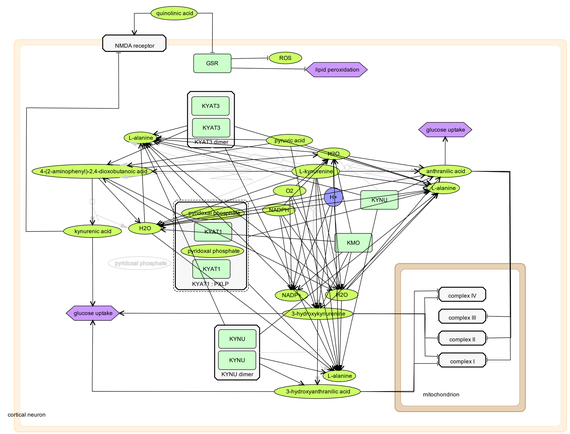

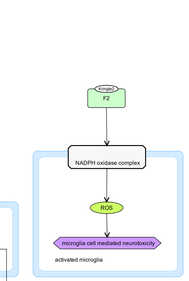

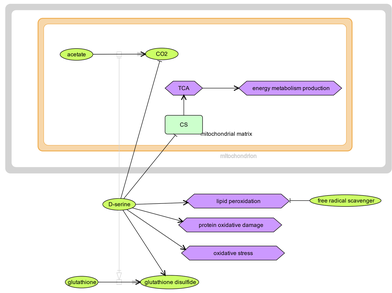

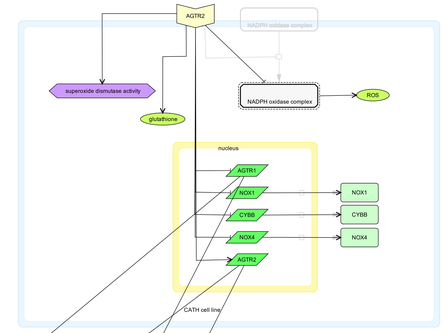

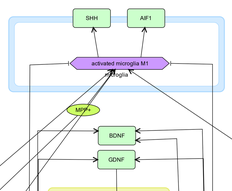

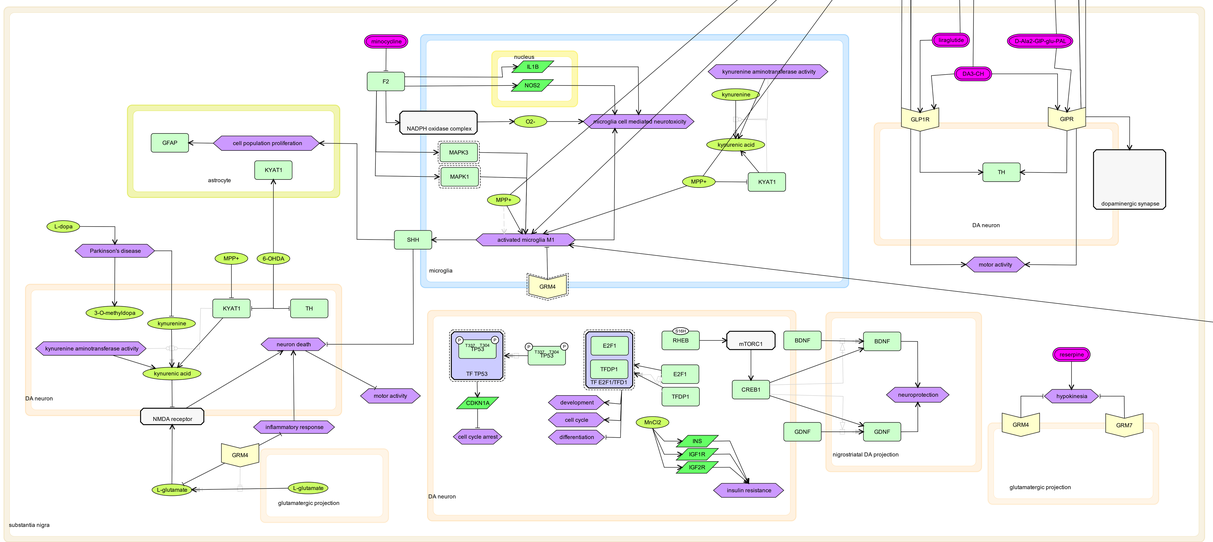

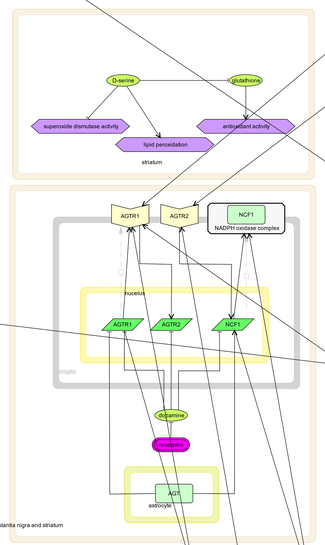

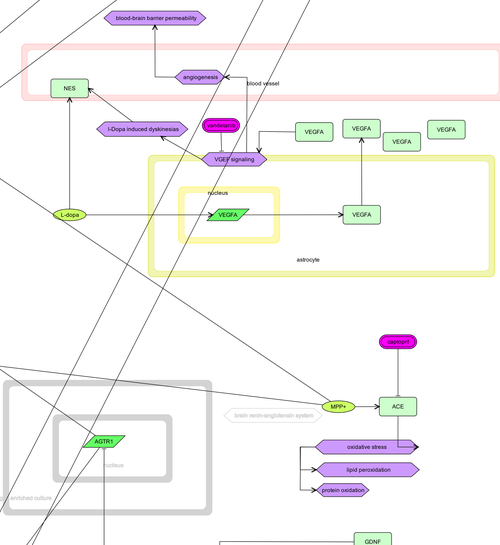

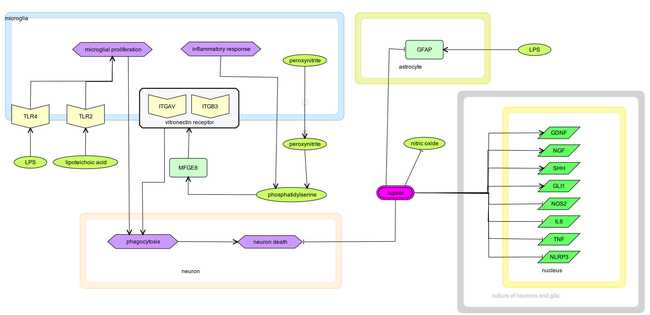

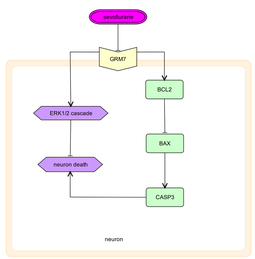

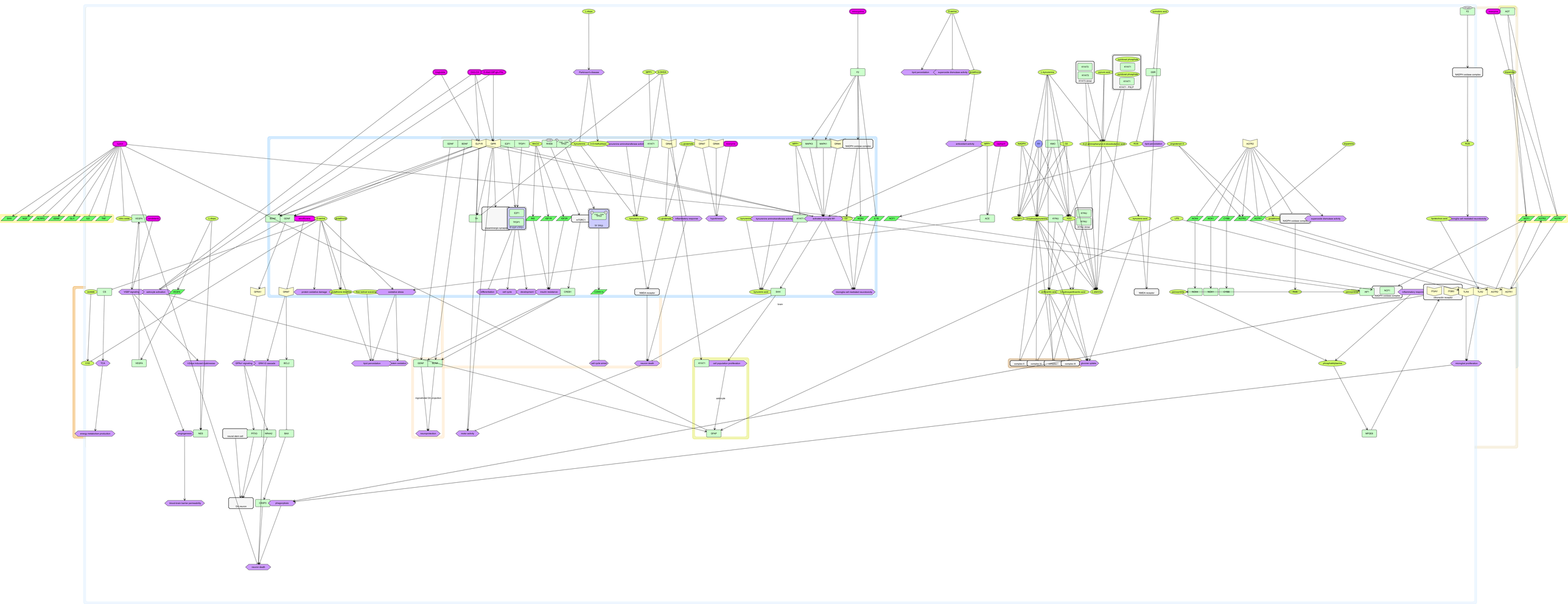

quinolinic acid
NMDA receptor
GSR
ROS
lipid peroxidation
KYAT3
KYAT3
KYAT3 dimer
L-alanine
pyruvic acid
H2O
glucose uptake
4-(2-aminophenyl)-2,4-dioxobutanoic acid
L-kynurenine
anthranilic acid
L-alanine
O2
H+
KYNU
pyridoxal phosphate
NADPH
kynurenic acid
H2O
KYAT1
KMO
pyridoxal phosphate
pyridoxal phosphate
KYAT1
KYAT1 : PXLP
NADP+
H2O
complex IV
complex III
complex II
complex I
glucose uptake
3-hydroxykynurenine
KYNU
KYNU
KYNU dimer
L-alanine
3-hydroxyanthranilic acid
mitochondrion
cortical neuron
SHH
AIF1
activated microglia M1
microglia
MPP+
BDNF
GDNF
astrocyte activation
GFAP
astrocyte
liraglutide
D-Ala2-GIP-glu-PAL
DA3-CH
GLP1R
GIPR
TH
dopaminergic synapse
DA neuron
motor activity
Kringle2
F2
NADPH oxidase complex
ROS
microglia cell mediated neurotoxicity
activated microglia
acetate
CO2
TCA
energy metabolism production
CS
mitochondrial matrix
mitochondrion
D-serine
lipid peroxidation
free radical scavenger
protein oxidative damage
oxidative stress
glutathione
glutathione disulfide
AGTR2
NADPH oxidase complex
superoxide dismutase activity
NADPH oxidase complex
ROS
glutathione
nucleus
AGTR1
NOX1
NOX1
CYBB
CYBB
NOX4
NOX4
AGTR2
CATH cell line
minocycline
F2
nucleus
IL1B
NOS2
kynurenine aminotransferase activity
NADPH oxidase complex
O2-
microglia cell mediated neurotoxicity
kynurenine
kynurenic acid
MAPK3
MAPK1
MPP+
KYAT1
MPP+
SHH
activated microglia M1
microglia
GRM4
GFAP
cell population proliferation
KYAT1
astrocyte
L-dopa
Parkinson's disease
MPP+
6-OHDA
3-O-methyldopa
kynurenine
KYAT1
TH
kynurenine aminotransferase activity
kynurenic acid
neuron death
motor activity
DA neuron
NMDA receptor
inflammatory response
GRM4
L-glutamate
L-glutamate
glutamatergic projection
substantia nigra
P
T337
T304
P
TP53
TF TP53
P
T337
T304
P
TP53
CDKN1A
cell cycle arrest
E2F1
TFDP1
TF E2F1/TFD1
E2F1
TFDP1
development
cell cycle
differentiation
S16H
RHEB
mTORC1
CREB1
MnCl2
INS
IGF1R
IGF2R
insulin resistance
DA neuron
BDNF
BDNF
GDNF
GDNF
neuroprotection
nigrostriatal DA projection
reserpine
hypokinesia
GRM4
GRM7
glutamatergic projection
D-serine
glutathione
superoxide dismutase activity
antioxidant activity
lipid peroxidation
striatum
AGTR1
AGTR2
NCF1
NADPH oxidase complex
nucleus
AGTR1
AGTR2
NCF1
microglia
dopamine
reserpine
AGT
astrocyte
substantia nigra and striatum
blood-brain barrier permeability
angiogenesis
blood vessel
NES
l-Dopa induced dyskinesias
vandetanib
VEGFA
VEGFA
VEGFA
VEGFA
VGEF signaling
nucleus
L-dopa
VEGFA
VEGFA
astrocyte
AGTR1
nucleus
microglia enriched culture
dopamine
captopril
MPP+
ACE
brain renin-angiotensin system
oxidative stress
lipid peroxidation
protein oxidation
GDNF
microglia
microglial proliferation
inflammatory response
peroxynitrite
TLR4
TLR2
ITGAV
ITGB3
vitronectin receptor
LPS
lipoteichoic acid
MFGE8
peroxynitrite
phosphatidylserine
phagocytosis
neuron death
neuron
GFAP
LPS
astrocyte
nitric oxide
lupeol
GDNF
NGF
SHH
GLI1
NOS2
IL6
TNF
NLRP3
nucleus
culture of neurons and glia
sevoflurane
GRM7
BCL2
ERK1/2 cascade
BAX
neuron death
CASP3
neuron
AGTR1
AGTR2
NCF1
NADPH oxidase complex
AGTR1
AGTR2
nucleus
NCF1
microglia
angiotensin II
GFRA1
GFRα1 signaling
NR4A2
PITX3
nucleus
ventral mesencephalon
neural stem cell
DA neuron
neuron
brain

L-dopa
minocycline
D-serine
quinolinic acid
Kringle2
F2
reserpine
AGT
liraglutide
DA3-CH
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KYAT1
KYAT1 : PXLP
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lupeol
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BDNF
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GRM4
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KYNU
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oxidative stress
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